

IMO(51st) – 2010

First Day

1. Determine all functions $f : R \rightarrow R$ such that the equality $f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor$ holds for all $x, y \in R$. (here $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z .)
2. Let I be the incentre of triangle ABC and let Γ be its circumcircle. Let the line AI intersect Γ again at D . Let E be a point on the arc BDC and F a point on the side BC such that $\angle BAF = \angle CAE < \frac{1}{2} \angle BAC$. Finally, let G be the mid point of the segment IF . Prove that the lines DG and EI intersect on Γ .
3. Let N be the set of positive integers. Determine all functions $g : N \rightarrow N$ such that $(g(m) + n)(m + g(n))$ is a perfect square for all $m, n \in N$.

Second Day

4. Let P be a point inside the triangle ABC . The lines AP , BP and CP intersect the circumcircle Γ of triangle ABC again at the points K , L and M respectively. The tangent to Γ at C intersect the line AB at S . Suppose that $SC = SP$. Prove that $MK = ML$.
5. In each of six boxes $B_1, B_2, B_3, B_4, B_5, B_6$ there is initially one coin. There are two types of operation allowed:
Type 1: Choose a nonempty box B_j with $1 \leq j \leq 5$. Remove one coin from B_j and add two coins to B_{j+1} .
Type 2: Choose a nonempty box B_k with $1 \leq k \leq 4$. Remove one coin from B_k and exchange the contents of (possibly empty) boxes B_{k+1} and B_{k+2} .
 Determine whether there is a finite sequence of such operations that results in boxes B_1, B_2, B_3, B_4, B_5 being empty and box B_6 containing exactly $2010^{2010^{2010}}$.
6. Let $a_1, a_2, a_3, \dots$ be a sequence of positive real numbers. Suppose that for some positive integer s , we have $a_n = \max\{a_k + a_{n-k} \mid 1 \leq k \leq n-1\}$ for all $n > s$. Prove that there exist positive integers ℓ and N with $\ell \leq s$ and such that $a_n = a_\ell + a_{n-\ell}$ for all $n \geq N$.
